Design Settings

Rainfall Methodology	FSR	Maximum Time of Concentration (mins)	30.00
Return Period (years)	1	Maximum Rainfall (mm/hr)	50.0
Additional Flow (%)	0	Minimum Velocity (m/s)	1.00
FSR Region	England and Wales	Connection Type	Level Soffits
M5-60 (mm)	20.000	Minimum Backdrop Height (m)	0.200
Ratio-R	0.400	Preferred Cover Depth (m)	1.200
CV	0.750	Include Intermediate Ground	✓
Time of Entry (mins)	5.00	Enforce best practice design rules	✓

Nodes

Name	Area (ha)	T of E (mins)	Cover Level (m)	Diameter (mm)	Easting (m)	Northing (m)	Depth (m)
1	0.019	5.00	102.675	1050	419006.192	186919.877	1.490
2	0.000	5.00	102.000	1050	419013.641	186948.365	1.045
3 Flow Control	0.000	5.00	101.546	2100	419024.083	186970.890	0.501
5	0.014	5.00	102.275	1050	419014.877	186930.145	1.180
6	0.030	5.00	102.300	1050	418999.660	186948.144	1.199
7	0.040	5.00	102.300	1050	419012.836	186943.199	1.293
8	0.000	5.00	101.900		419016.369	186962.628	0.750
9	0.000	5.00	104.000		419031.204	186977.105	3.050

Links

Name	US Node	DS Node	Length (m)	ks (mm) / n	US IL (m)	DS IL (m)	Fall (m)	Slope (1:X)	Dia (mm)	T of C (mins)	Rain (mm/hr)
1.000	1	5	13.448	0.600	101.485	101.395	0.090	149.4	225	5.10	50.0
1.001	5	7	13.213	0.600	101.395	101.307	0.088	150.1	225	5.25	50.0
2.000	6	7	14.073	0.600	101.401	101.307	0.094	149.7	225	5.15	50.0
1.002	7	2	5.238	0.600	101.307	101.255	0.052	100.7	300	5.43	50.0
1.003	2	8	14.514	0.600	101.255	101.150	0.105	138.2	300	5.50	50.0
1.004	8	3 Flow Control	11.303	0.600	101.150	101.045	0.105	107.6	300	5.64	50.0
1.005	3 Flow Control	9	9.452	0.600	101.045	100.950	0.095	99.5	150	5.80	50.0

Name	Vel (m/s)	Cap (l/s)	Flow (l/s)	US Depth (m)	DS Depth (m)	Σ Area (ha)	Σ Add Inflow (l/s)	Pro Depth (mm)	Pro Velocity (m/s)
1.000	1.067	42.4	2.6	0.965	0.655	0.019	0.0	37	0.592
1.001	1.064	42.3	4.5	0.655	0.768	0.033	0.0	49	0.692
2.000	1.066	42.4	4.1	0.674	0.768	0.030	0.0	47	0.679
1.002	1.566	110.7	14.0	0.693	0.445	0.103	0.0	71	1.079
1.003	1.335	94.4	14.0	0.445	0.450	0.103	0.0	78	0.965
1.004	1.515	107.1	14.0	0.450	0.201	0.103	0.0	73	1.057
1.005	1.007	17.8	14.0	0.351	2.900	0.103	0.0	100	1.112

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Pipeline Schedule

Link	Length (m)	Slope (1:X)	Dia (mm)	Link Type	US CL (m)	US IL (m)	US Depth (m)	DS CL (m)	DS IL (m)	DS Depth (m)
1.000	13.448	149.4	225	Circular_Default Sewer Type	102.675	101.485	0.965	102.275	101.395	0.655
1.001	13.213	150.1	225	Circular_Default Sewer Type	102.275	101.395	0.655	102.300	101.307	0.768
2.000	14.073	149.7	225	Circular_Default Sewer Type	102.300	101.401	0.674	102.300	101.307	0.768
1.002	5.238	100.7	300	Circular_Default Sewer Type	102.300	101.307	0.693	102.000	101.255	0.445
1.003	14.514	138.2	300	Circular_Default Sewer Type	102.000	101.255	0.445	101.900	101.150	0.450
1.004	11.303	107.6	300	Circular_Default Sewer Type	101.900	101.150	0.450	101.546	101.045	0.201
1.005	9.452	99.5	150	Circular_Default Sewer Type	101.546	101.045	0.351	104.000	100.950	2.900

Link	US Node	Dia (mm)	Node Type	MH Type	DS Node	Dia (mm)	Node Type	MH Type
1.000	1	1050	Manhole	Adoptable	5	1050	Manhole	Adoptable
1.001	5	1050	Manhole	Adoptable	7	1050	Manhole	Adoptable
2.000	6	1050	Manhole	Adoptable	7	1050	Manhole	Adoptable
1.002	7	1050	Manhole	Adoptable	2	1050	Manhole	Adoptable
1.003	2	1050	Manhole	Adoptable	8		Junction	
1.004	8		Junction		3 Flow Control	2100	Manhole	Adoptable
1.005	3 Flow Control	2100	Manhole	Adoptable	9		Junction	

Manhole Schedule

Node	Easting (m)	Northing (m)	CL (m)	Depth (m)	Dia (mm)	Connections		Link	IL (m)	Dia (mm)
1	419006.192	186919.877	102.675	1.490	1050		0	1.000	101.485	225
2	419013.641	186948.365	102.000	1.045	1050		1	1.002	101.255	300
							0	1.003	101.255	300
3 Flow Control	419024.083	186970.890	101.546	0.501	2100		1	1.004	101.045	300
							0	1.005	101.045	150
5	419014.877	186930.145	102.275	1.180	1050		1	1.000	101.395	225
							0	1.001	101.395	225
6	418999.660	186948.144	102.300	1.199	1050		0	2.000	101.401	225
7	419012.836	186943.199	102.300	1.293	1050		1	2.000	101.307	225
							2	1.001	101.307	225
							0	1.002	101.307	300
8	419016.369	186962.628	101.900	0.750			1	1.003	101.150	300
							0	1.004	101.150	300

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Manhole Schedule

Node	Easting (m)	Northing (m)	CL (m)	Depth (m)	Dia (mm)	Connections	Link	IL (m)	Dia (mm)
9	419031.204	186977.105	104.000	3.050		1	1.005	100.950	150

Simulation Settings

Rainfall Methodology	FSR	Drain Down Time (mins)	1440
FSR Region	England and Wales	Additional Storage (m³/ha)	0.0
M5-60 (mm)	20.000	Check Discharge Rate(s)	✓
Ratio-R	0.400	1 year (l/s)	3.9
Summer CV	0.750	30 year (l/s)	11.0
Winter CV	0.840	100 year (l/s)	14.6
Analysis Speed	Normal	Check Discharge Volume	✓
Skip Steady State	✓	100 year 1440 minute (m³)	

Storm Durations

15	30	60	120	180	240	360	480	600	720	960	1440

Return Period (years)	Climate Change (CC %)	Additional Area (A %)	Additional Flow (Q %)
1	0	0	0
30	0	0	0
100	40	0	0

Pre-development Discharge Rate

Site Makeup	Greenfield	Growth Factor 30 year	2.40
Greenfield Method	IH124	Growth Factor 100 year	3.19
Positively Drained Area (ha)	0.043	Betterment (%)	0
SAAR (mm)	675	QBar	
Soil Index	4	Q 1 year (l/s)	
SPR	0.47	Q 30 year (l/s)	
Region	6	Q 100 year (l/s)	
Growth Factor 1 year	0.85		

Pre-development Discharge Volume

Site Makeup	Greenfield	Return Period (years)	100
Greenfield Method	FSR/FEH	Climate Change (%)	0
Positively Drained Area (ha)	0.043	Storm Duration (mins)	1440
Soil Index	4	Betterment (%)	0
SPR	0.47	PR	
CWI	101.482	Runoff Volume (m³)	

Node 3 Flow Control Online Hydro-Brake® Control

Flap Valve	x	Objective	(HE) Minimise upstream storage
Replaces Downstream Link	✓	Sump Available	✓
Invert Level (m)	101.045	Product Number	CTL-SHE-0103-5000-1200-5000
Design Depth (m)	1.200	Min Outlet Diameter (m)	0.150
Design Flow (l/s)	5.0	Min Node Diameter (mm)	1200

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Node 3 Flow Control Depth/Area Storage Structure

Base Inf Coefficient (m/hr)	0.00000	Safety Factor	2.0	Invert Level (m)	101.045
Side Inf Coefficient (m/hr)	0.00000	Porosity	1.00	Time to half empty (mins)	96

Depth (m)	Area (m²)	Inf Area (m²)	Depth (m)	Area (m²)	Inf Area (m²)
0.000	163.0	0.0	0.700	381.0	0.0

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Results for 1 year Critical Storm Duration. Lowest mass balance: 99.77%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
15 minute winter	1	10	101.523	0.038	2.7	0.0329	0.0000	OK
15 minute winter	2	9	101.345	0.090	14.1	0.0778	0.0000	OK
180 minute winter	3 Flow Control	132	101.100	0.055	3.6	9.6426	0.0000	OK
15 minute winter	5	10	101.445	0.049	4.6	0.0428	0.0000	OK
15 minute winter	6	10	101.448	0.047	4.2	0.0410	0.0000	OK
15 minute winter	7	10	101.388	0.081	14.3	0.0703	0.0000	OK
15 minute winter	8	9	101.332	0.182	14.1	0.0000	0.0000	OK
15 minute summer	9	1	100.950	0.000	0.6	0.0000	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
15 minute winter	1	1.000	5	2.6	0.490	0.062	0.0730	
15 minute winter	2	1.003	8	14.1	0.858	0.150	0.4538	
180 minute winter	3 Flow Control	Hydro-Brake®	9	1.5				15.3
15 minute winter	5	1.001	7	4.5	0.475	0.107	0.1278	
15 minute winter	6	2.000	7	4.2	0.446	0.098	0.1333	
15 minute winter	7	1.002	2	14.1	0.949	0.128	0.0848	
15 minute winter	8	1.004	3 Flow Control	19.0	1.660	0.177	0.2549	

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Results for 30 year Critical Storm Duration. Lowest mass balance: 99.77%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
15 minute winter	1	10	101.546	0.061	6.6	0.0525	0.0000	OK
15 minute winter	2	8	101.391	0.136	34.9	0.1176	0.0000	OK
120 minute winter	3 Flow Control	90	101.157	0.112	11.1	20.5390	0.0000	OK
15 minute winter	5	10	101.473	0.078	11.3	0.0678	0.0000	OK
15 minute winter	6	10	101.476	0.075	10.4	0.0650	0.0000	OK
15 minute winter	7	10	101.447	0.140	35.3	0.1208	0.0000	OK
15 minute winter	8	7	101.377	0.227	34.8	0.0000	0.0000	OK
15 minute summer	9	1	100.950	0.000	2.4	0.0000	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
15 minute winter	1	1.000	5	6.5	0.627	0.154	0.1403	
15 minute winter	2	1.003	8	34.8	1.168	0.369	0.6296	
120 minute winter	3 Flow Control	Hydro-Brake®	9	3.8				32.1
15 minute winter	5	1.001	7	11.2	0.591	0.264	0.2519	
15 minute winter	6	2.000	7	10.3	0.552	0.243	0.2633	
15 minute winter	7	1.002	2	34.9	1.130	0.316	0.1634	
15 minute winter	8	1.004	3 Flow Control	35.6	2.053	0.332	0.3246	

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Results for 100 year +40% CC Critical Storm Duration. Lowest mass balance: 99.77%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
15 minute winter	1	10	101.566	0.080	11.9	0.0697	0.0000	OK
15 minute winter	2	7	101.458	0.203	63.1	0.1761	0.0000	OK
120 minute winter	3 Flow Control	98	101.247	0.202	20.7	40.0098	0.0000	FLOOD RISK
15 minute winter	5	10	101.529	0.133	20.6	0.1156	0.0000	OK
15 minute winter	6	10	101.526	0.125	18.9	0.1084	0.0000	OK
15 minute winter	7	10	101.512	0.205	63.5	0.1772	0.0000	OK
15 minute winter	8	7	101.424	0.274	63.1	0.0000	0.0000	OK
15 minute summer	9	1	100.950	0.000	4.2	0.0000	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
15 minute winter	1	1.000	5	11.8	0.653	0.278	0.2506	
15 minute winter	2	1.003	8	63.1	1.417	0.669	0.8584	
120 minute winter	3 Flow Control	Hydro-Brake®	9	4.7				59.1
15 minute winter	5	1.001	7	20.2	0.645	0.477	0.4127	
15 minute winter	6	2.000	7	18.5	0.595	0.435	0.4266	
15 minute winter	7	1.002	2	63.1	1.299	0.570	0.2567	
15 minute winter	8	1.004	3 Flow Control	63.9	2.293	0.597	0.3867	

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